

Final Exam

3 hours; closed-book and closed-notes; calculators are not permitted. There are 12 required questions (200 points) + 4 optional questions. **The decision on whether or not you pass the course will be based only on the required questions.**

Required Questions

1. (24 pts) Find $f'(x)$.

(a) $f(x) = \frac{1}{x^4} - 3 \cos(2x + 1), \quad x \neq 0$

(b) $f(x) = x^5 \ln(1 + e^{2x})$

(c) $f(x) = (1 + \sin^2 x)^{\sqrt{x}}, \quad x > 0$

2. (24 pts) Evaluate the limit, or explain why it does not exist.

(a) $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{e^{x^2} - 1}$

(b) $\lim_{x \rightarrow 0} \left(1 - \frac{x}{2}\right)^{\frac{3}{x}}$

(c) $\lim_{x \rightarrow \infty} x(e^{\frac{2}{x}} - 1)$

3. (10 pts) Let $f(x) = x^3 e^{-x^4}$. Find $f^{(27)}(0)$ (the value at 0 of the 27th derivative of f).

4. (16 pts) Evaluate each indefinite integral.

(a) $\int (\ln x)^2 dx$

(b) $\int \frac{1}{\sqrt{x} + 2} dx$

5. (16 pts) Find $F'(x)$. You may leave un-evaluated integrals in your answer.

(a) $F(x) = x^2 \int_x^1 e^{t^2} dt$

(b) $F(x) = \int_0^{x^3} \cos(t^2 x^2) dt$

6. (18 pts) Determine whether or not each improper integral converges. Give a brief explanation. (Complete proofs are not required.)

(a) $\int_0^{\infty} \frac{x^2}{(1 + \sqrt{x})^5} dx$

(b) $\int_e^{\infty} \frac{(\ln x)^{10}}{x^3} dx$

(c) $\int_0^{\infty} \frac{x^5 e^{-\sqrt{x}}}{1 + x^4} dx$

7. (12 pts) Suppose that

$$z = e^{2x} + y^3.$$

Suppose further that x and y are smooth functions of t such that when $t = 1$ we have

$$x = 0, \quad \frac{dx}{dt} = 2, \quad \frac{d^2x}{dt^2} = -1, \quad y = 1, \quad \frac{dy}{dt} = 3, \quad \frac{d^2y}{dt^2} = -2.$$

Find the value of $\frac{d^2z}{dt^2}$ when $t = 1$.

8. (16 pts) Write an equivalent double integral with the order of integration reversed.

(a) $\int_0^3 \int_0^{6-2x} f(x, y) dy dx$

(b) $\int_0^1 \int_{e^y}^e f(x, y) dx dy$

9. (16 pts) Evaluate each double integral.

(a) $\int_0^1 \int_0^{\sqrt{x}} x^3 y dy dx$

(b) $\int_0^1 \int_{\sqrt{x}}^1 e^{y^3} dy dx$

10. (16 pts) Solve each of the following initial value problems.

(a) $y'(t) = \frac{t^2(1 + y(t)^3)}{3y(t)^2}; \quad y(0) = 1$

(b) $y'(t) + \frac{2}{t}y(t) = 1; y(1) = 2$ (We want a solution for $t > 0$ here.)

11. (16 pts) Find the general solution of each of the following.

(a) $y''(t) + 8y'(t) + 16y(t) = 0$

(b) $t^2y''(t) - 2y(t) = 0$

12. (16 pts) Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & -1 \\ 1 & 2 & 1 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & -1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

(a) Find $\det(A)$.

(b) Find the rank of B .

Optional Questions

13. (10 pts) Evaluate

$$\int_1^2 \frac{\ln t}{1+t} dt + \int_1^{\frac{1}{2}} \frac{\ln t}{1+t} dt.$$

14. (10 pts) Find all continuously differentiable functions satisfying

$$f(x)^2 = 100 + \int_0^x [f(t)^2 - f(t)^4 + (f'(t))^2] dt$$

on some open interval containing 0. (The interval on which the solution is defined could be different for different solutions.)

15. (10 pts) Find the general solution of

$$t^4y''(t) + 2t^2y'(t) + (1 - 2t)y(t) = 0, \quad t > 0.$$

16. (10 pts) Let N be a given positive integer and consider the $N \times N$ matrix A given by

$$A = \begin{pmatrix} N & 1 & 1 & \cdot & \cdot & \cdot & 1 \\ 1 & N & 1 & \cdot & \cdot & \cdot & 1 \\ 1 & 1 & N & \cdot & \cdot & \cdot & 1 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & 1 & \cdot & \cdot & \cdot & 1 & N \end{pmatrix}.$$

(The diagonal entries of A are all equal to N and all of the other entries are equal to 1.) Find the eigenvalues of A .